Week 1 Lab

Vivado Installation and Top-Down Hierarchical Design

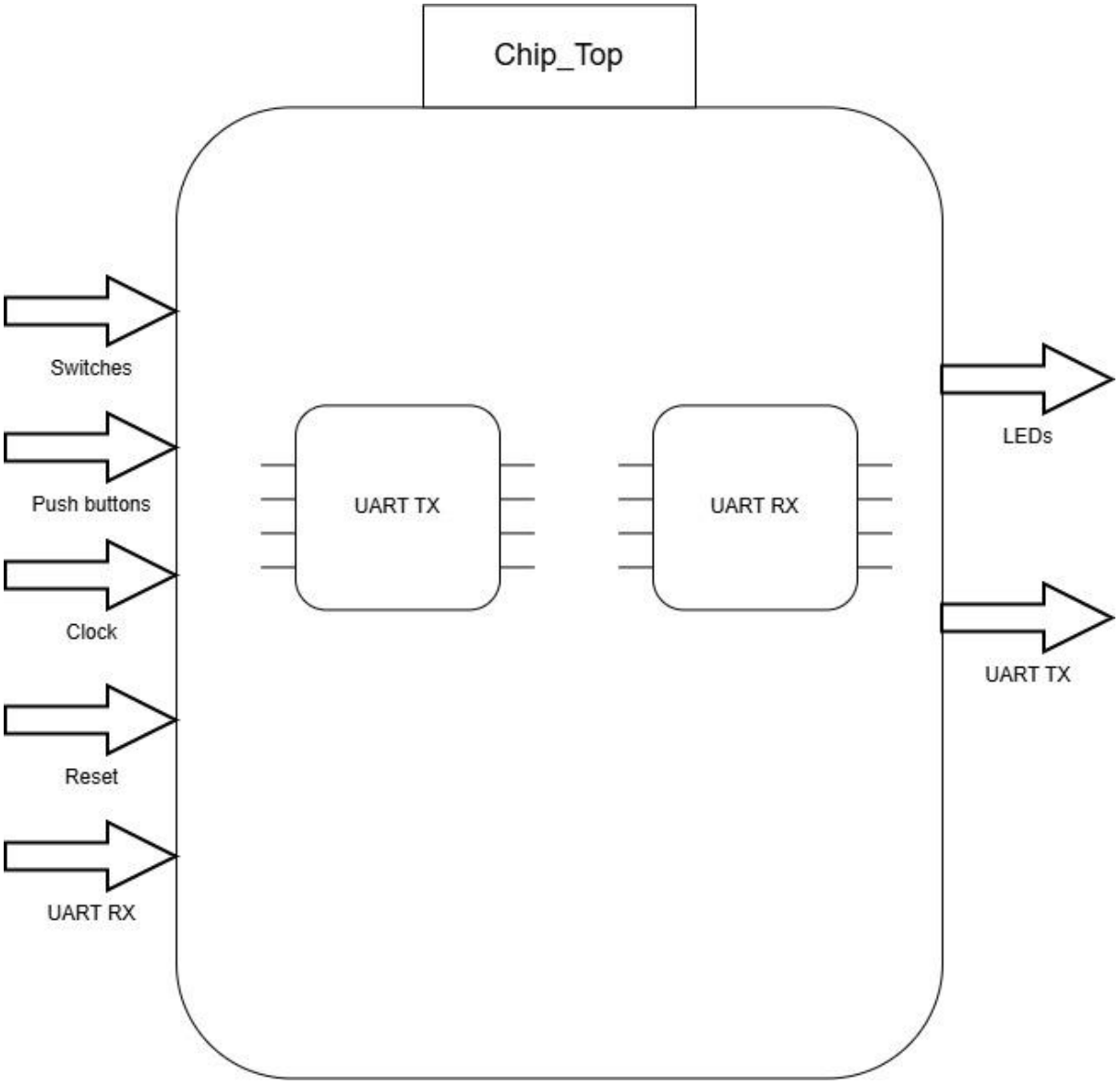
Name – Mohamad Mashal

Objective

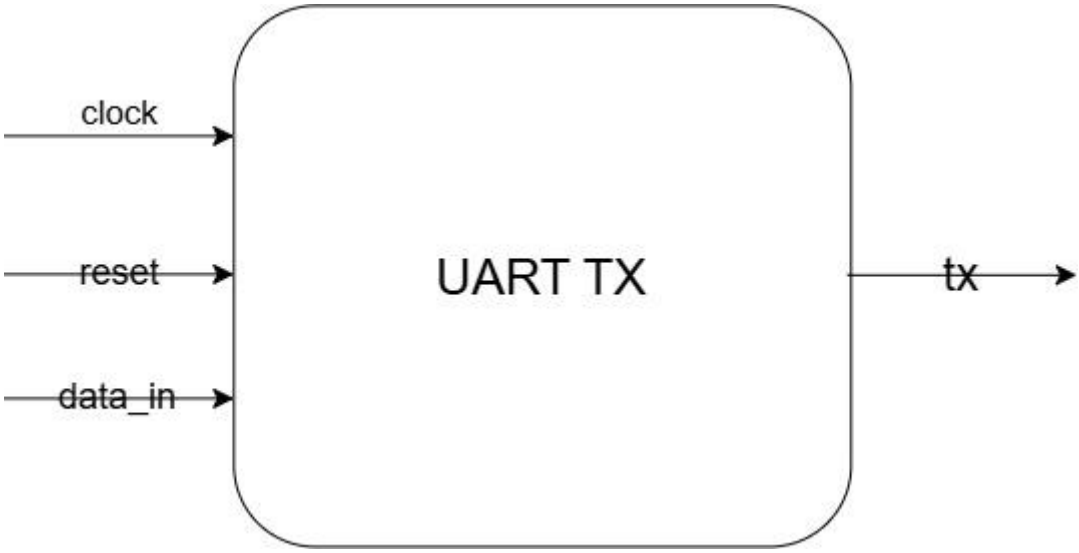
The goal of this lab is to design and implement multiple stubs for our Core project. We will have a chip_top stub that interacts with switches, leds, clocks, and UART interface. We will see how a top-down Architecture works before we go down to u-arch:

1. Methodology differences: Why Top-Down works better for architectural work.

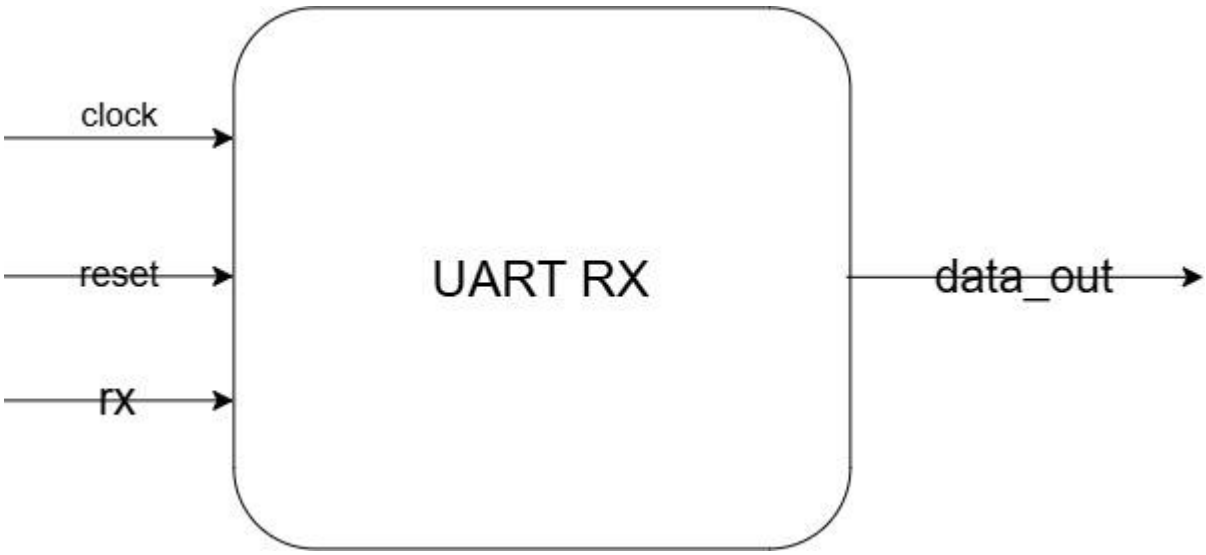
2. Creating stubs, instantiations and linting the design: Basic steps in architectural design and working with the Vivado suite.



Port Name	Direction	Type	Description
switches	input	wire	Switch inputs
push_buttons	input	wire	Push-button inputs
clock	input	wire	System clock
reset	input	wire	System reset
uart_rx	input	wire	Serial communication input
leds	output	wire	LED outputs
uart_tx	output	wire	Serial communication output

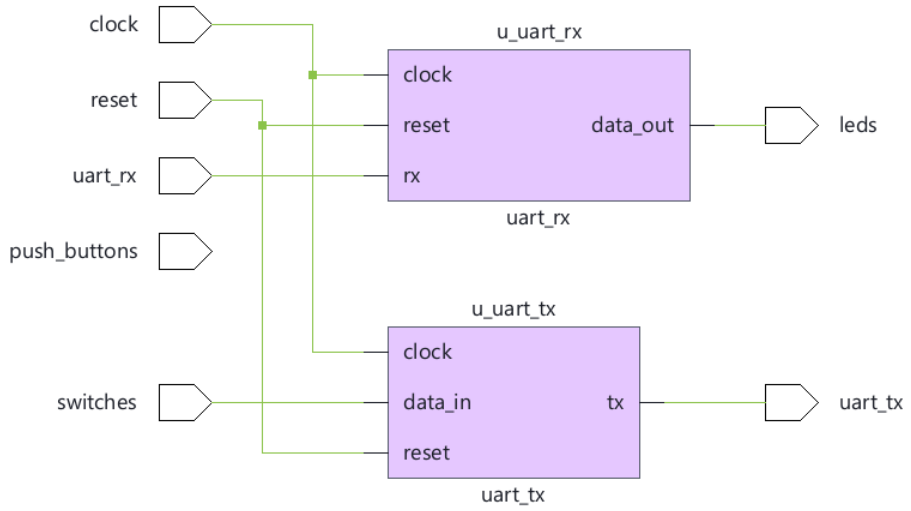


Port Name	Direction	Type	Description
clock	input	wire	System clock
reset	input	wire	System reset
data_in	input	wire	Data to be transmitted
tx	output	wire	Serial transmission output



Port Name	Direction	Type	Description
clock	input	wire	System clock
reset	input	wire	System reset
rx	input	wire	Serial reception input
data_out	output	wire	Received data

Vivado Elaborated Design



The diagram shows the chip_top hierarchy and the connections to the UART TX and UART RX modules. The push_buttons input is currently unconnected.